# FDO2-G2

## Optical Oxygen Sensor

DATA SHEET

O2

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## Optical Oxygen Sensor

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**The FDO2-G2 is manufactured by:**

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# TABLE OF CONTENT

# 1 INTRODUCTION

The **FDO2-G2** is an optical oxygen sensor for gas measurements based on luminescence quenching of a sensor dye. The dye is excited with red light, and the properties of the resulting luminescence are measured in the near infrared. The presence of molecular oxygen quenches the luminescence, changing its intensity and lifetime fully reversibly.

This principle is very robust. In most applications it shows no interferences to other relevant gases, has a very low drift, and the sensor is fully solid-state. It virtually does not deplete over time, unlike galvanic oxygen sensors with their limited shelf life. Optics and electronics are hermetically sealed from the measured gas. For typical indoor environmental applications, up to 10-year operating life is expected.

The sensor comes with a factory calibration. If required, the user can perform a simple 1-point calibration at ambient air. The **FDO2-G2** features build in temperature compensation and a digital interface that provides oxygen partial pressure values. No additional signal conditioning is necessary. A mounting thread and a robust locking connector allow easy installation.

## Features

- High-accuracy measurement
- Low drift
- Factory calibrated
- Long life
- Fast response ($t_{63}$<2s)
- Digital output of oxygen partial pressure
- Temperature compensation
- Low power consumption
- Optoelectronics hermetically sealed from the measured medium
- Lead free, ROHS compliant

## Potential Applications

- Incubators
- Oxygen concentrators
- Inert gas processing chambers (glove boxes)
- Exhaust gas measurement
- Inert gas monitoring
- Portable equipment
- Monitoring fruit storage and transport
- Biogas applications
- Oxygen level monitoring in buildings

# 2 SPECIFICATIONS

**Analytical Performance**
Given for factory calibration. Units of % O2 given for 1013 mbar ambient air pressure. If not otherwise indicated given for default measurement mode M=2 (refer to command #SETM in section 4)

| | | |
|---|---|---|
| **Measuring Range**<br>Typical<br>Maximum (not specified) | hPa<br>0-1000 hPa<br>0-2000 hPa | % $O_2$<br>0-100 % $O_2$ (gas) |
| **Accuracy 10 °C – 40 °C**<br>at 10 hPa/1 % $O_2$<br>at 200 hPa/20 % $O_2$<br>at 1000 hPa/100 % $O_2$ | <br>±0.2 hPa<br>±5 hPa<br>±20 hPa | <br>±0.02 % $O_2$<br>±0.5 % $O_2$<br>±2 % $O_2$ |
| **Accuracy -10 °C – 60 °C**<br>at 10 hPa/1 % $O_2$<br>at 200 hPa/20 % $O_2$ | <br>±1 hPa<br>±10 hPa | <br>±0.1 % $O_2$<br>±1 % $O_2$ |
| **Resolution**<br>at 10 hPa/1 % $O_2$<br>at 200 hPa/20 % $O_2$<br>at 1000 hPa/100 % $O_2$ * | <br>±0.1 hPa<br>±1 hPa<br>±5 hPa * | <br>±0.01 % $O_2$<br>±0.1 % $O_2$<br>±0.5 % $O_2$ * |
| **Detection Limit** | 0.1 hPa | 0.01 % $O_2$ |
| **Response Time (t63)** | <2 s | |
| **Drift** | typ. <1 % $O_2$ / year ** | |
| **Max. number of measurements** | >500 million *** | |
| **Lifetime** | typ. >5 years *** | |
| **Warm-up Time** | 3 min (reduced accuracy during warm-up) | |
| **Internal atmospheric pressure sensor measurement range** | 300 – 1100 mbar (measured through the venting capillary at the backside of the housing) | |

* given for factory calibration and high-resolution settings (M=4 adjusted by command #SETM, refer to section 4)
** at 21 % $O_2$, 25 °C, 1013 mbar ambient gas pressure, protected from direct sunlight. The drift can be significantly increased after the exposure to elevated temperature >60 °C or to specific chemicals (refer to section 3).
*** at 21 % $O_2$, 25 °C, 1013 mbar ambient gas pressure, protected from direct sunlight.

Note, the FDO2 outputs oxygen partial pressure $pO_2$ (hPa). The unit volume percent oxygen (% $O_2$) is given here only for convenience. If the air pressure *p*Air at the oxygen sensing membrane is identical to the air pressure at the venting capillary on the backside of the housing, then the measurement of the internal pressure sensor can be used for converting $pO_2$ into units of % $O_2$ by the formula:

$$\text{volume percent oxygen } [\%\ O_2] = 100 \times pO_2\,[\text{hPa}] \,/\, p\text{Air}\,[\text{hPa}]$$

| **Environment** | |
|---|---|
| **Temperature range during operation** | -10 to 60 °C **** |
| **Temperature range during storage** | -40 °C to 60 °C **** |
| **Humidity** | Backside: Non-condensing<br>Sensing Membrane: Dew point must not be within the membrane |
| **Maximum absolute pressure** | 20 bar |
| **Maximum differential pressure** | 3 bar |

**** Exposure to temperatures >60 °C might lead to increased drift of the oxygen measurement within the next weeks.

| **Interface** | |
|---|---|
| **Supply Voltage** | 3.3 – 5.0 V DC (absolute maximum rating 5.5 V DC) |
| **Standby / Peak Currents** | ca. 8 mA / 40 mA |
| **Energy Consumption per Measurement** | ca. 1-2 mAs |
| **Communication Interface** | 3.0 V UART (absolute maximum rating 3.3 V DC) |
| **Connector** | Molex 560020-0420 |

| Mechanical | |
|---|---|
| **Dimensions** | Ø 28,5 mm x 28 mm |
| **Weight** | 10,5 g |
| **Mounting Thread** | M16 x 1 |
| **Housing Material** | Polycarbonate |
| **Conformity to RoHS directive** | RoHS compliant, lead free |

# 3 CROSS-SENSITIVITY AND CHEMICAL COMPATIBILITY

The following table shows the compatibility and possible cross sensitivities to some important chemical substances at a given concentration range. A "✓" under "OK" indicates compatibility. "Cross-Sensitivity" indicates that the oxygen measurement is influenced by this substance. "Damage" indicates that this substance might physically damage the FDO2-G2.

**Applicability, Cross-Sensitivity and Chemical Compatibility**

| Substance | Concentration | OK | Cross-Sensitivity | Damage | Comment |
|---|---|---|---|---|---|
| Moisture | 0 – 100 % | ✓ | | | |
| $CH_4$ | <20 % | ✓ | | | |
| $Cl_2$ | | | X | X | |
| CO | <20 % | ✓ | | | |
| $CO_2$ | <20 % | ✓ | | | |
| $H_2S$ | <1 % | ✓ | | | |
| NO | <1 % | ✓ | | | 1 |
| $NO_2$ | | | X | X | 2 |
| $N_2O$ | <1 % | ✓ | | | |
| Inorganic acids/ inorganic bases | <1 % | ✓ | | | |

| | | | | | |
|---|---|---|---|---|---|
| Methanol, Ethanol, Isopropanol, Formic acid, Acetic acid; Acetone | <0.05 %v | ✓ | | | 3 |
| Methanol, Ethanol, Isopropanol, Formic acid, Acetic acid; Acetone | >0.05 %v | | X | | 4 |
| Other volatile organic compounds | | | X | X | 5 |

Comments:

1. NO may form $NO_2$ in presence of oxygen.
2. Ca. 5-10 times more sensitive to $NO_2$ than to oxygen. Slow degradation over time.
3. 0.1 %v in gas corresponds approximately to the vapor pressure above a 0.5-1 % solution in water at 25 °C.
4. Recalibration after conditioning at constant substance levels might be possible.
5. Exposure to EtO (e.g. for sterilization) will cause increased drift. Recalibration after exposure is possible.
6. Can result in erroneous oxygen readings and significantly enhanced drift. Interference depends on the compound. Substances with high vapor pressure or high reactivity are expected to be more problematic.

# 4 COMMUNICATION INTERFACE

The communication interface is a standard UART (3.0 V levels, max. 3.3 V). The default settings of the UART after power-up are:

- 19200 Baud Rate, 8 Data Bits, 1 Stop Bit, no Parity, no Handshake

The module needs ca. 1 sec. power up time before it responds to commands. If the module comes with an USB-adapter-cable, the “Simple FDO2 Logger” software (for PC with operation system “Windows”) installs automatically a virtual serial port driver (COM-port), which is directly connected to the internal UART interface of the module. The virtual COM-port can be accessed by standard COM-port libraries in any programming language under Windows.

## 4.1 General Definitions

Every command sent from the “master” to the device must be terminated by a single carriage return (0x0D), or by a carriage return followed by a line feed (0x0D, 0x0A). This termination is indicated in the following by the symbol “↵”. Values (numbers) are separated with spaces from each other and the command header. The command header is a unique string of characters. If the command with all values could be successfully interpreted by the device, the command is echoed **after completion of the requested task**, appended with requested values and terminated by **a single carriage return (0x0D) only**. Otherwise, the response begins with an error header followed by an error code. Italic letters represent placeholders for numbers, which are transmitted as ASCII-Strings (human readable). **The absolute maximum range of all values transmitted in the communication protocol is from -2147483648 to +2147483647**. The only exception is the command #IDNR which can return higher values.

## 4.2 Detecting communication errors and optional CRC

The protocol supports two levels of communication error handling. The first level is realized simply by the fact, that the device always echoes the complete command as it was received from the master. This way the master can compare the echo with the originally sent command. If there is a difference, then the master should send the command again.

However, the echoing of the command does not allow detecting potential communication errors within the returned values added by the device to the original command. For this purpose, it is possible to enable a cyclic redundancy check (CRC) based on the CRC16 definition as used for MODBUS protocols (CRC-16-IBM). The CRC is enabled by the command #CRCE (see below). Now every answer from the device to the master is terminated by the string ":␣*C*↵" where *C* is the CRC16 checksum represented as a decimal ASCII-string (human readable). The CRC is calculated for all ASCII-characters (unsigned bytes) from the very beginning of the returned string until the character just before the ":" proceeding the CRC16 checksum. Please note, that also spaces (ASCII code 32) are included in the CRC calculation.

The reminder of this document shows only examples with **disabled** CRC. So, the returned string from the device is only terminated by "↵", i.e. a single carriage return (0x0D).

## 4.3 Commands

**Read Device Information:** #VERS ↵

**Response** #VERS *D N R S* ↵

Returns general information about the device: *D* represents the device id, which is always *D*=8 for any **FDO2-G2** module, *N* returns the number of oxygen channels which is *N*=1 for the **FDO2-G2** module. *R* represents the firmware revision (e.g. *R*=328 designates firmware revision 3.28). *S* is a byte number indicating which sensor types are available. The bits refer to the following sensor types: bit 0: oxygen, bit 1: temperature within housing, bit 2: pressure within housing, bit 3: relative humidity within housing. If *S*=15 (default) then the module is equipped with all these sensor types.

**Read Unique ID Number:** #IDNR ↵

**Response** #IDNR *N* ↵

Returns an identification number *N* being unique for each single device, which is an unsigned 64 bit integer. Therefore, the decimal ASCII representation of this number can be up to 20 chars in length. Please note, that the unique ID number is not equivalent to the serial number off the device.

**Measure Oxygen and return the results:** #MOXY ↵

**Response** #MOXY *O T S* ↵

Measures (1) the temperature, (2) the oxygen partial pressure (pO2), (3) the total pressure + humidity WITHIN the sensor housing, (4) ambient light and returns the results:

O: Oxygen partial pressure, signed 32 bit integer in units of $10^{-3}$ hPa
(O = 203456 corresponds to 203.456 hPa, typ. range 0-210 hPa)

T: Temperature, signed 32 bit integer in units of m°C
(T = 17892 corresponds to 17.892 °C
T = -1965 corresponds to -1.965 °C, typ. range 0 – 50 °C)

S: Status (unsigned 32 bit integer) with warning and error bits, where bit(x)=1 means:
bit(0): WARNING, the detector amplification was automatically reduced in order to avoid saturation of the detector. The oxygen reading is still valid. This might happen at low temperatures together with low oxygen values causing high luminescent intensities of the oxygen indicator. Or the sensor might be exposed to excessive ambient light (e.g. sun light).
bit(1): FATAL ERROR, oxygen sensor signal intensity too low (<20 mV)
bit(2): FATAL ERROR, oxygen sensor signal or ambient light too high
bit(3): FATAL ERROR, oxygen reference signal intensity too low (<20 mV)
bit(4): FATAL ERROR, oxygen reference signal or ambient light too high (>2400 mV)
bit(5): FATAL ERROR, failure of the temperature sensor

bit(6): reserved
bit(7): WARNING, the humidity within the sensor housing is >90 %RH. This might lead to fatal electronic problems and therefore to a failure of the oxygen measurement.
bit(8): reserved
bit(9): ERROR, failure of the pressure sensor within the sensor housing, this has no direct influence on the measured oxygen partial pressure
bit(10): ERROR, failure of the humidity sensor within the sensor housing, this has no direct influence on the measured oxygen partial pressure

IMPORTANT: Under normal operation the status should be always S=0 or S=1 . In all other cases the oxygen (O) and the temperature (T) readings are or can be faulty, although this command is still returning values for O and T. It is on the users authority to check the status for each single measurement, in order to detect a faulty sensor operation. **Especially the error bits marked with „FATAL ERROR“ will lead to incorrect oxygen readings.**

## Measure Partial Pressure Oxygen and Return Additional Raw Data:

#MRAW ↲

**Response**

#MRAW *O T S D I A P H* ↲

Measures (1) the temperature, (2) the oxygen partial pressure (pO2), (3) the total pressure + humidity WITHIN the sensor housing, (4) ambient light and returns the results including raw data:

O: refer to the command #MOXY
T: refer to the command #MOXY
S: refer to the command #MOXY
D: the phase shift „dphi", signed 32 bit integer in units of m° (millidegrees)
(e.g. D = 24385 corresponds to 24.385°, typ. range 5°-60°)
I: the signal intensity of the oxygen sensor, signed 32 bit integer in units of µV
(e.g. I = 124072 corresponds to 123.072 mV, typ. range 100-500 mV)
A: the ambient light entering the sensor, signed 32 bit integer in units of µV
(e.g. A = 12792 corresponds to 12.792 mV, typ. range 0 - ca. 100 mV)
P: the ambient pressure at the BACKSIDE of the module, signed 32 bit integer in units of µbar
(e.g. P = 999734 corresponds to 999.734 mbar, typ. range 900-1100mbar)
H: the relative humidity inside the housing in units of m%RH
(e.g. H = 40365 corresponds to 40.365 %RH, typ. range 10-90 %RH)

**dphi (D).** The fundamental raw data of the oxygen measurement is the phase shift „dphi" (D) which is together with the temperature (T) internally converted into the oxygen reading (O).

**Signal intensity (I).** This represents the measured luminescent intensity of the oxygen sensor. The signal intensity can be used for checking the quality of the sensor. Please note, that the signal intensity is highly dependent on the actual oxygen level. Therefore, measured signal intensities can be only compared with each other, if they have been measured at similar oxygen levels. Typical signal intensities are 200-400 mV at oxygen levels of ca. 200 hPa at room temperature.

**Ambient light (A).** Immediately before each oxygen measurement, the device measures also the ambient light entering the oxygen detector. Considerable amounts of ambient light might enter the optoelectronics, if the sensor is exposed e.g. to direct sun light. This might lead to a saturation of the optical detector and to an invalid oxygen measurement (see parameter Status S). As a thumb of rule the sum of ambient light (A) and signal intensity (I) should not exceed ca. 2000 mV. If it exceeds this value, the sensor must be shaded from the external light source.

**Ambient pressure (P).** This gives the ambient air pressure measured at the BACKSIDE of the housing (i.e. the side where the electrical connector is positioned).

**Relative humidity (H).** The relative humidity measured inside the housing including the optoelectronics. The inner volume of the housing is connected via a thin channel to the BACKSIDE of the housing (connector side). Therefore, the humidity will be in the long run influenced by the humidity present at the BACKSIDE of the housing. A rel. humidity > 90 % will lead to a warning in the status (S).

**Indicate Logo:** #LOGO ↵

**Response** #LOGO ↵

The status LED of the device flashes 4 times (total duration ca. 1 second). This command can be used to identify which device belongs to which interface port, if several devices are operated in parallel.

**Enable or disable CRC checksum:** Note WARNING below! * #CRCE K ↲

**Response** #CRCE K ↲

K = 0: Disable the CRC checksum (default)
K = 1: Enable the CRC checksum
The current status is automatically saved in the flash memory, so the status is persistent even after a power cycle.

**Read User Memory:** #RDUM R N ↲

**Response** #RDUM R N $Y_1$ … $Y_N$ ↲

The device offers a user memory of altogether 64 signed 32bit integer numbers (range -2147483648 to 2147483647) which is located in the flash memory and is therefore retained even after power cycles. This read command returns *N* (*N*=1…64) consecutive values $Y_1$ … $Y_N$ from the user memory starting at the user memory address *R* (*R*=0…63). Note, that *N*+*R* must be <=64.

**Write User Memory:** #WRUM R N $Y_1$ … $Y_N$ ↲

**Response** #WRUM R N $Y_1$ … $Y_N$ ↲

This command writes the *N* (*N*=1…64) values $Y_1$ … $Y_N$ consecutively starting at the user memory address *R* (*R*=0…63). Note, that *N*+*R* must be <=64.
IMPORTANT: This command must be used economically, because the flash memory is designed for typ. max. 20000 flash cycles. Each time this command is executed, it will trigger a flash cycle.

**Set Baud Rate:** #BAUD R ↲

**Response** #BAUD R ↲

Changes the current baud rate of the UART interface to R. Supported baud rates are 1200, 2400, 4800, 9600, 14400, *19200*, 28800, 38400, 56000, 57600, 115200. The default baud rate after a power cycle is always 19200. If the command was successfully interpreted, the response string is sent back by using the old baud rate and afterwards the baud rate is changed. For highest reliability it is recommended to use the default baud rate of 19200.

**Calibrate Oxygen Sensor at 0 % oxygen:** Note WARNING below! * #CALO ↲

**Response** #CALO ↲

**Execution of this command will result in a loss of the factory calibration!**
Calibrates the oxygen sensor at 0 % oxygen, giving the lower calibration point of a two-point oxygen calibration. The sensor should be measuring (with the application specific sample interval) for at least 5 min. under steady state conditions, before this command is executed. This command needs up to 10 seconds for completion, before the response is returned. The new calibration is automatically stored in the flash memory and kept even after a power cycle.

Note: The zero calibration point is very stable. Only for highly accurate measurements close to 0 % $O_2$, a calibration of this point should be considered. For most applications a 1-point-calibration (e.g. at ambient air) based on the #CAHI command is sufficient.

**Calibrate Oxygen Sensor at given partial pressure oxygen:** Note WARNING below! * #CAHI P ↲

**Response** #CAHI P ↲

**Execution of this command will result in a loss of the factory calibration!**
P: Oxygen partial pressure in the given calibration standard, signed 32 bit integer in units of $10^{-3}$ hPa (P = 203456 corresponds to 203.456 hPa)
Calibrates the oxygen sensor at a given calibration standard (not equal 0 % oxygen!), giving the upper calibration point of a two-point oxygen calibration. The sensor should be measuring (with the application specific sample interval) for at least 5 min. under steady state conditions, before this command is executed. This command needs up to 10 seconds for completion, before the response is returned. The new calibration is automatically stored in the flash memory and kept even after a power cycle.

**Enable/Disable Broadcast Mode:** Note WARNING below! * #BCST T ↲

**Response** #BCST T ↲

Enables or disables the broadcast mode, and stores the actual status always in the flash memory, so the actual status is persistent even after a power cycle.
T = 0: Broadcast mode is disabled (default).
T = 100..10000: Broadcast mode is enabled with the time interval of T ms. So time intervals between 0.1 s and 10 s are possible. In broadcast mode the modules triggers itself periodic measurements and returns the results via the UART interface as given by the answer string of the #MRAW command. Note, that the time interval is only an approximated value, as the duration of the actual oxygen measurement and the transmission of the UART response must be added to this interval.
Note, if a command is sent to the module, while the module is transmitting a broadcast message, then the broadcast message might be interrupted at any position by the answer of the requested command. Therefore, it is recommended to disable the broadcast mode (command #BCST 0↲) before sending other commands to the module.

**Set flash duration mode:** Note WARNING below! * #SETM M ↲

### Response

#SETM *M* ↲

Selects the flash duration mode M (values from 0 to 4) and stores it in the flash memory, so the actual status is persistent even after a power cycle. By adjusting this mode, the user can select the best compromise between low drift and low measurement noise for specific applications. M=0 corresponds to lowest drift with decreased measurement resolution. M=4 corresponds to highest measurement resolution (lowest measurement noise) accompanied by increased drift. The default value is M=2.
M = 0: lowest drift, a complete measurement takes maximum approx. 150 ms
M = 1: lower drift, a complete measurement takes maximum approx. 150 ms
M = 2: factory setting (default) a complete measurement takes maximum 150 ms
M = 3: lower noise (higher resolution), a complete measurement takes maximum approx. 300 ms
M = 4: lowest noise (highest resolution), a complete measurement takes maximum approx. 300 ms

* WARNING:

The commands **#CALO, #CAHI, #CRCE, ‘SETM and #BCST** automatically trigger that the internal configuration and calibration registers of the FDO2 are saved into its internal flash memory ("flash cycle"), in order to make the changes persistent even after power cycles. **Note, the FDO2 supports max. ca. 20,000 flash cycles triggered by these commands. If the power supply is interrupted during a flash cycle, then the FDO2 flash memory might be corrupted.**

SAFETY MEASURES

- **Use the commands #CALO, #CAHI, #CRCE, ‘SETM and #BCST economically**, i.e. do not use them more frequently than really needed.
- **Ensure a stable power supply** when executing these commands.
- **Always evaluate the response sent back by these commands**. After these commands have been executed, the FDO2 will send back the identical command string as an acknowledgement. If you do not receive this answer, the flash memory might be corrupted.

## 4.4 Error Codes

If a command triggered an internal error or could not be interpreted correctly an error string is sent back instead of the default echo. An error string always begins with the error header "#ERRO" followed by a space character **(0x20)**, a negative error *E* number (starting with a "minus" character **(0x2D)**), and a carriage return **(0x0D)**.

*Table 1: Error Codes*

#ERRO E ↵

| E | Error Type | Description |
|---|---|---|
| -1 | *General* | A non-specific error occurred. |
| -21 | *UART Parse* | An error occurred while parsing the command string. The last command should be repeated. Check the command syntax! |
| -22 | *UART Rx* | The command string was not received correctly (e.g. device was not ready, last command was not terminated correctly). Repeat the last command. |
| -23 | *UART Header* | The command header could not be interpreted correctly (must contain only characters from A-Z). Repeat the last command. |
| -26 | *UART Request* | The command header does not match any of the supported requests. |
| -60 | *Invalid Calibration* | The performed oxygen calibration is outside the expected range. The calibration is discarded |

Any other error code not listed here corresponds to a potentially fatal error. Replace the sensor and contact PyroScience.

# 5 RELEASED FIRMWARE REVISIONS CHANGES

The **FDO2-G2** was initially released with Firmware version 3.55 (05/2025)

## Changes in firmware revision 3.57 (03/2026)

- The new firmware revision supports now modified electronics as indicated in product change notification PCN-FDO2-G2-001 (added low-pass filter in amplification pathway). Backward compatibility to former electronics is ensured.
- The issue of the errata "Increased Noise at Specific Temperatures" was resolved (refer to 6.1)

# 6 ERRATA

## 6.1 Increased Noise at Specific Temperatures

### Affected Modules

All modules with firmware revision 3.55.

### Description

The oxygen reading of some modules may exhibits increased noise typ. at two specific temperatures. The exact temperatures vary from module to module, but they are typ. located between 8°C and 15°C. This increased noise is observable within ca. +-0.1°C at these specific temperatures. For example, a module might show this increased noise within the temperature intervals of 14.1 - 14.3°C and 9.7 - 9.9°C. This increased noise is indeed not significantly higher than the basic noise level. It is comparable to the specified resolution as shown in the following table:

| Partial pressure oxygen pO2 (hPa) | Official specifications Resolution (hPa) | Increased RMS* noise (hPa) at two specific temperatures |
|---|---|---|
| 10 | +- 0.1 | ca. +- 0.2 |
| 200 | +- 1 | ca. +- 1 |
| 1000 | +- 5 | ca. +- 6 |

* root mean square

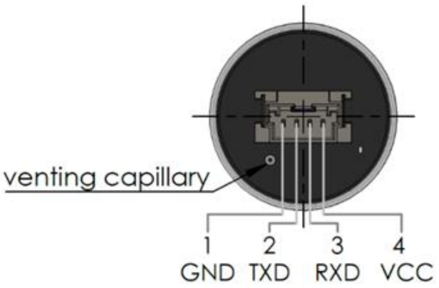
### Workaround

The customer may use the command #SETM in order to adapt the general noise level suitable for the respective application.

### Resolution

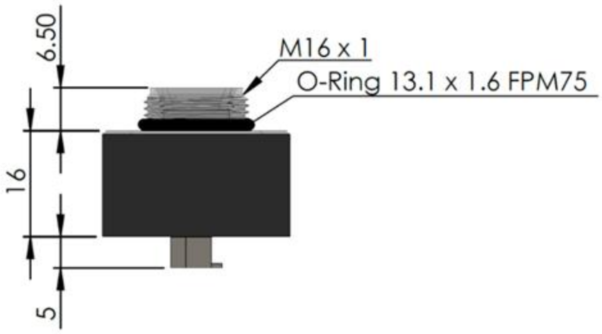
This issue was resolved with a hardware change in the signal amplification pathway and firmware revision 3.57 (refer to Product change notification PCN-FDO2-G2-001).

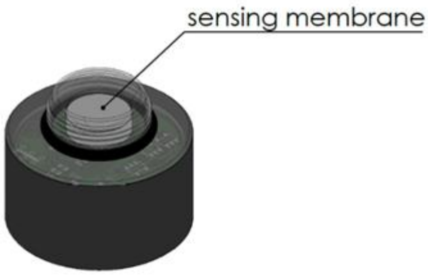
# 7 MECHANICAL DIMENSIONS AND ELECTRICAL INTERFACE




| Pin | Name | Function | Description |
|---|---|---|---|
| 1 | GND | Power | Ground |
| 2 | TXD | Digital output<br>3.0 V levels | Data transmission line of the UART interface |
| 3 | RXD | Digital input<br>3.0 V levels<br>(max. 3.3V tolerant) | Data receive line of the UART interface |
| 4 | VCC | Power | Power supply<br>min. 3.3 VDC<br>max. 5.0 VDC |

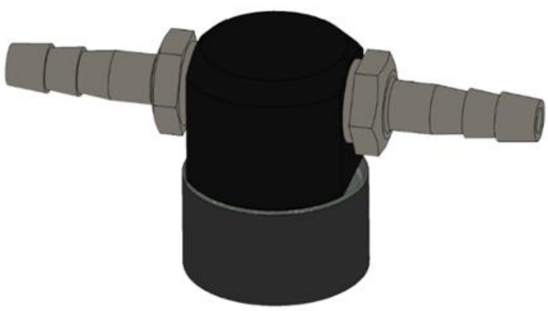
Connector: Molex 560020-0420





flow-through cell available as accessory



All units given in millimeter (mm).

# 8 WARNINGS

Note, this product does not come with any medical or safety-related certification. It is the user's responsibility to verify the product's suitability for such applications.

This product is not intended and not sold for aerospace applications, unless separately acknowledged by PyroScience. Please contact us regarding potential aerospace applications.

Avoid all sources of ignition especially if the sensors are used in pure oxygen or oxygen enriched atmospheres.

The data contained in this document is for guidance only. Customers should test under their own conditions, to ensure that the sensors are suitable for their own requirements.

# CONTACT

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